

UCI Online Retail

Business Data Analysis — Executive Insights

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Executive Summary

Four signals define the business story

£10.64M

Valid transaction revenue

£10.11M

Clean physical-product revenue

4,338

Known customers

£1.50M

November 2011 peak

KEY FINDINGS

Seasonality: November 2011 was the highest observed month at £1.50M.

Customer concentration: the top 26% of known customers generated 80% of known-customer revenue; Gini = 0.716.

Geographic concentration: the United Kingdom represented 84.59% of revenue and 90.0% of Champions.

Retention signal: 1,188 customers were classified as Potential Loyalists / At Risk; their historical revenue baseline was approximately £689K.

Core message: revenue is substantial, but concentration and customer recency deserve management attention.

Data Integrity & Critical Decisions

The final dataset is traceable from raw transactions to physical-product analysis

541,909

Raw transactions

524,878

Valid sales rows

522,728

Physical-product rows

£0.00

Revenue reconciliation difference

KEY QUALITY DECISIONS

Missing CustomerID: 24.93% of raw records are treated as Unknown for customer-level analysis.

Exact duplicates: 5,268 rows removed before analytical calculations.

Extreme outlier isolated: PAPER CRAFT , LITTLE BIRDIE — 80,995 units and £168,469.60.

Operational items separated from merchandise analysis: £362,103.47.

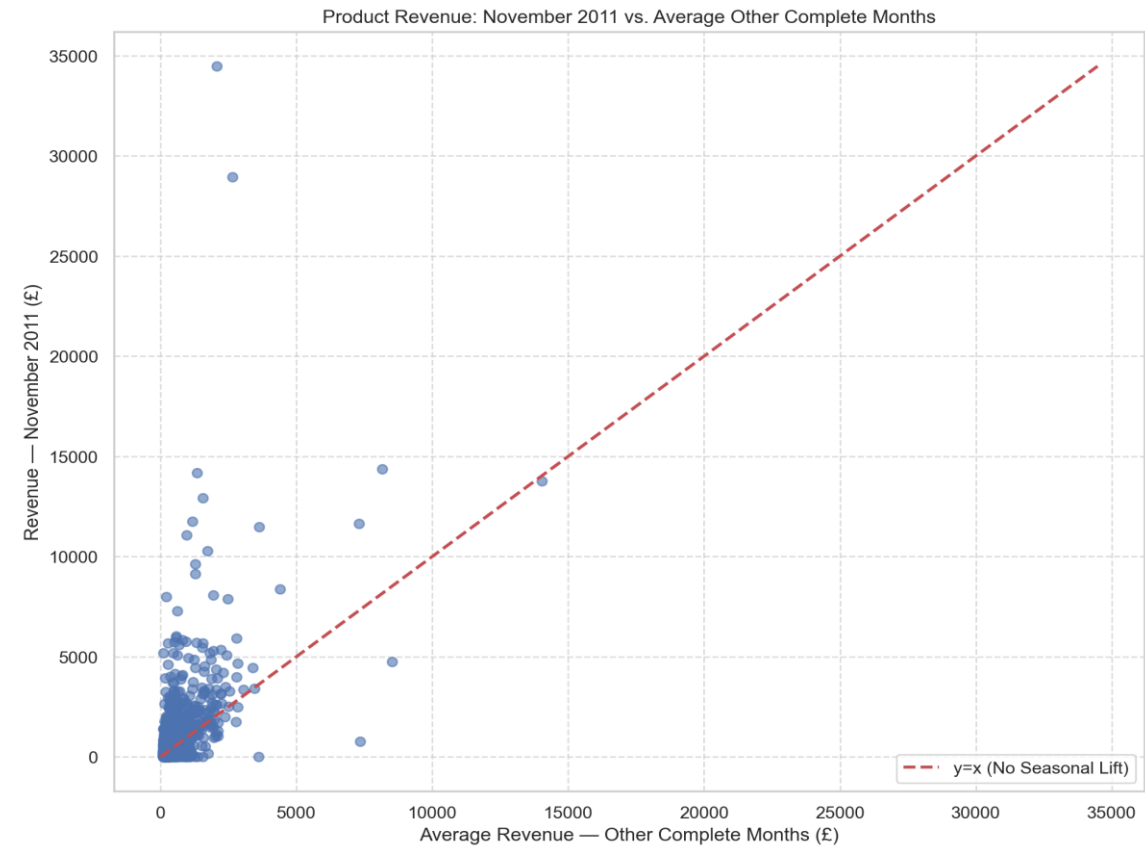
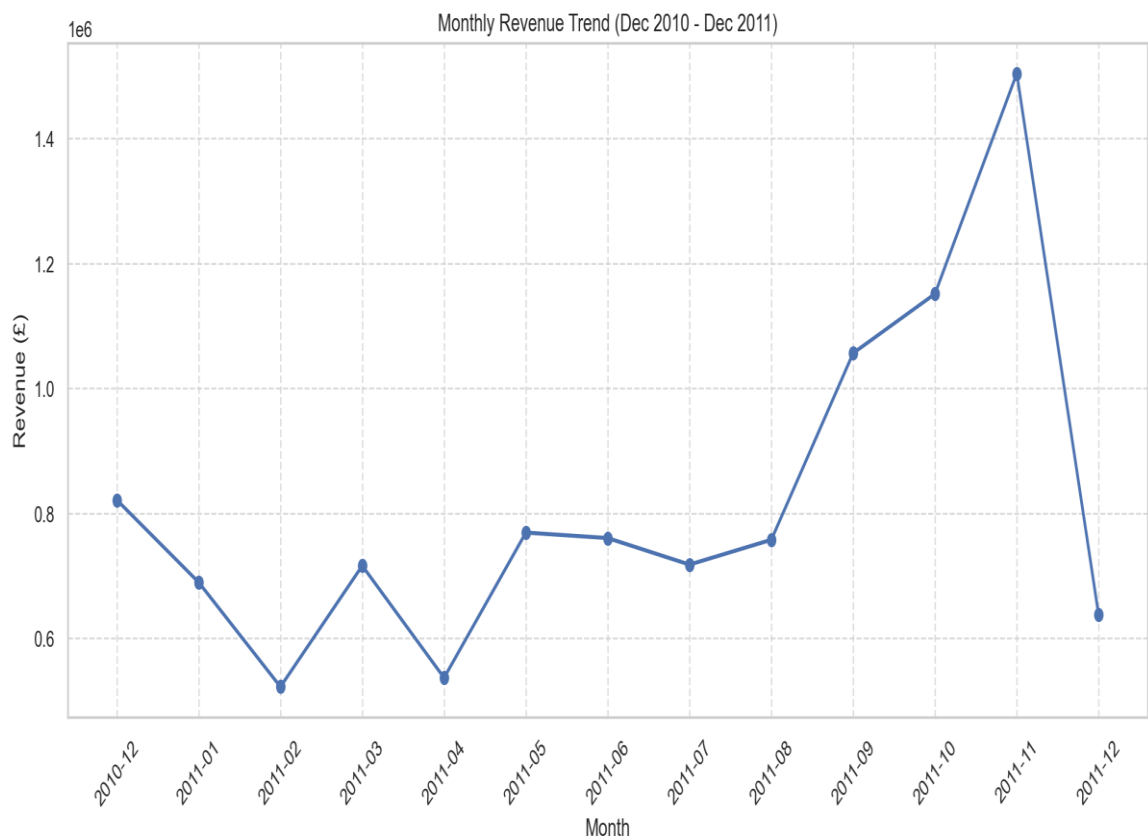
Final physical-product dataset: 522,728 rows and £10,111,537.73.

December 2011 is incomplete: the dataset ends on 9 December, so it is not treated as a full-month observation.

Validation result: row difference = 0 and revenue difference = £0.00.

Key Finding 1 — November Revenue Peak

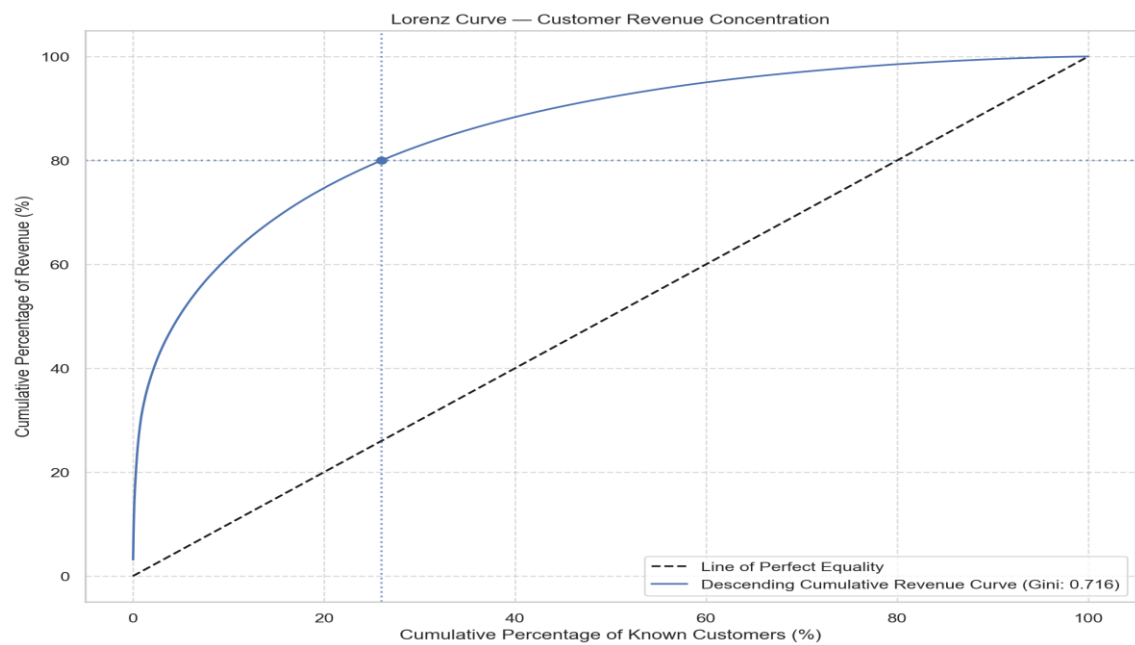
The November spike is accompanied by a clear change in product mix



November 2011: £1,503,866.78
RABBIT NIGHT LIGHT generated £34,478.40 with a seasonal lift of £32,409.02.

Key Finding 2 — Customer Concentration & RFM

Revenue is concentrated, while customer behaviour reveals four distinct groups



26%

Top customer share

80%

Known-customer revenue generated

0.716

Gini coefficient

2,073

At Risk + Lost customers

Champions

1,243 customers

Recency: 18 days
Frequency: 10.0
Monetary: £5,465

Loyal Customers

1,022 customers

Recency: 55 days
Frequency: 3.2
Monetary: £1,176

Potential Loyalists / At Risk

1,188 customers

Recency: 102 days
Frequency: 1.6
Monetary: £580

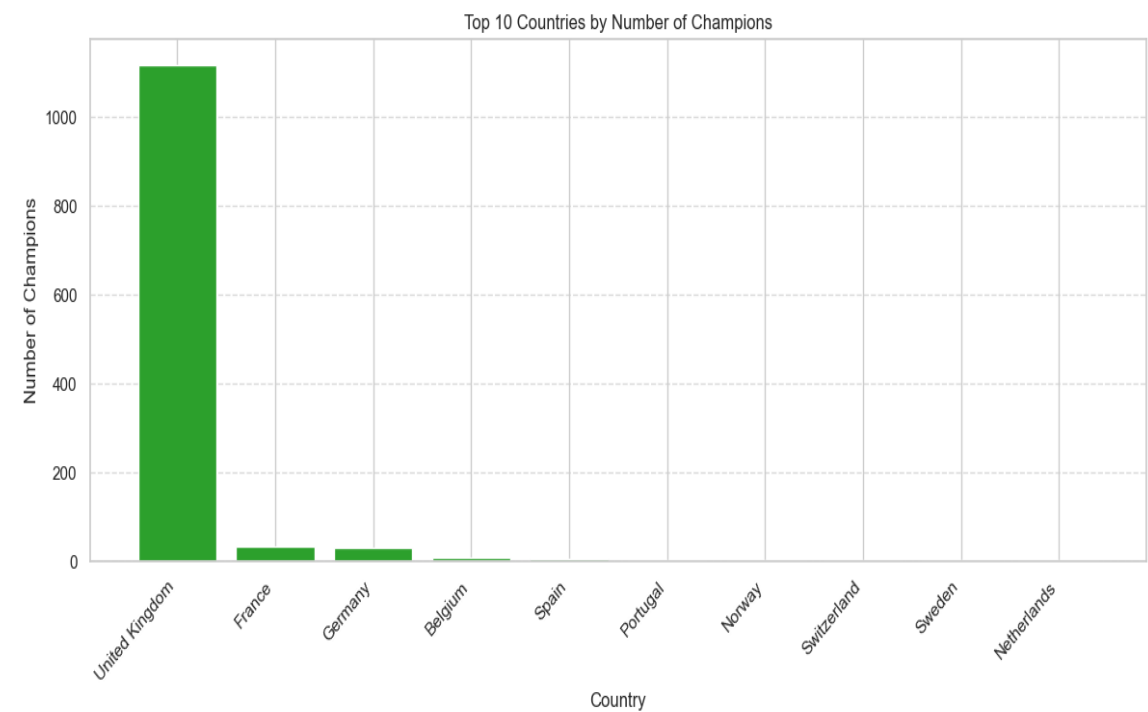
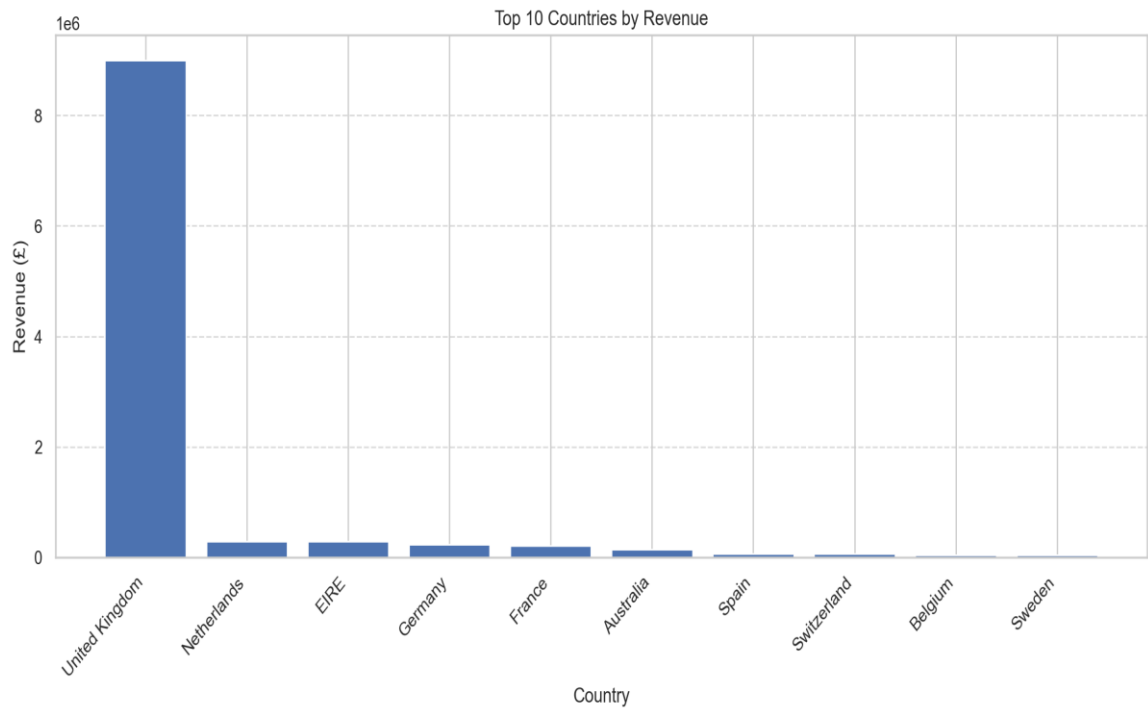
Lost / Hibernating

885 customers

Recency: 223 days
Frequency: 1.1
Monetary: £228

Key Finding 3 — Geographic Concentration

The United Kingdom dominates both revenue and the Champion customer base



84.59%

United Kingdom revenue share

1,119

UK Champions

35

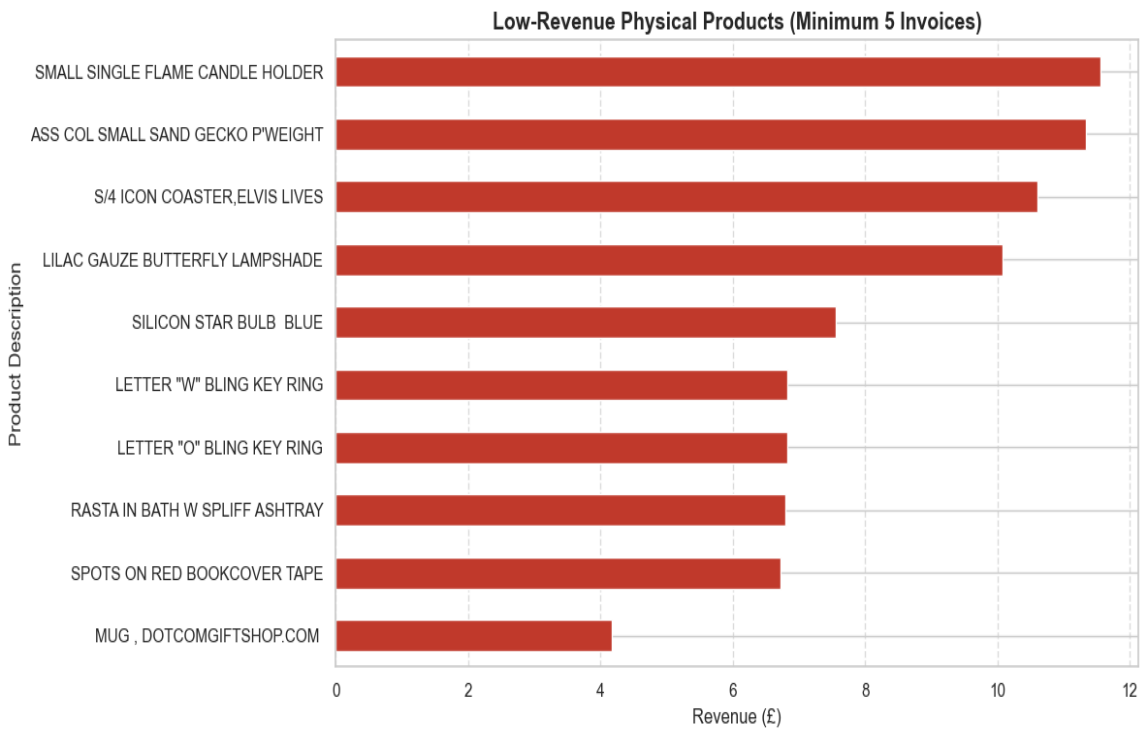
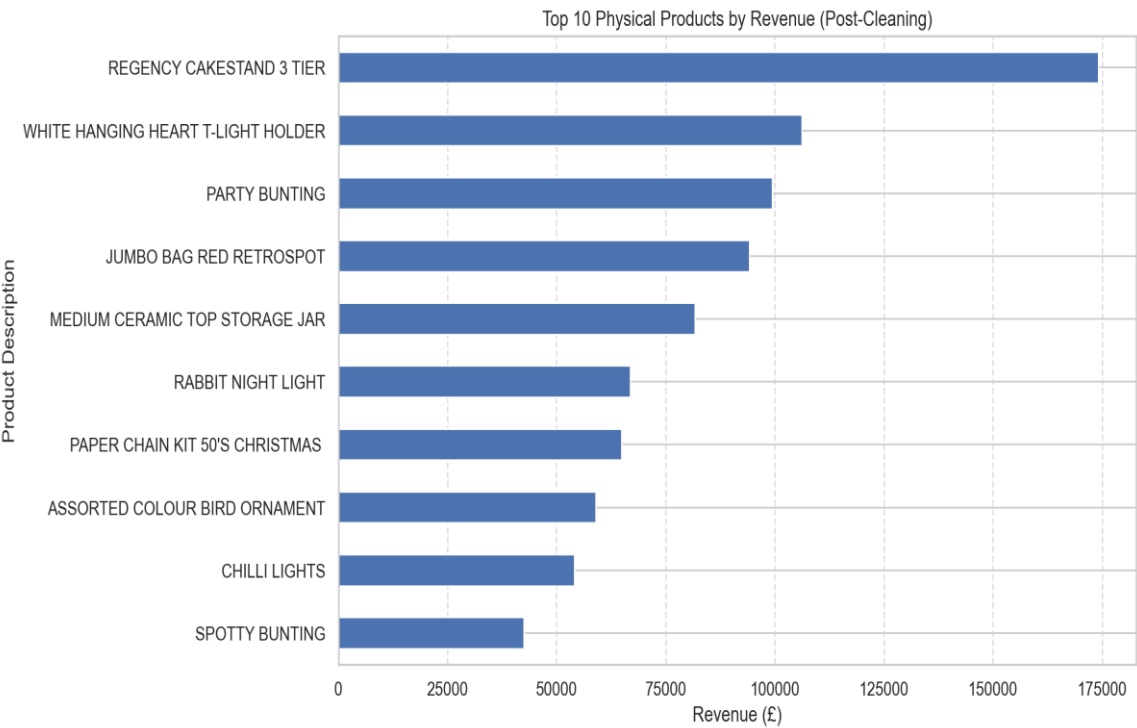
France Champions

33

Germany Champions

Product Portfolio — Winners & Low-Revenue Tail

Strong sellers coexist with a small group of very low-revenue products



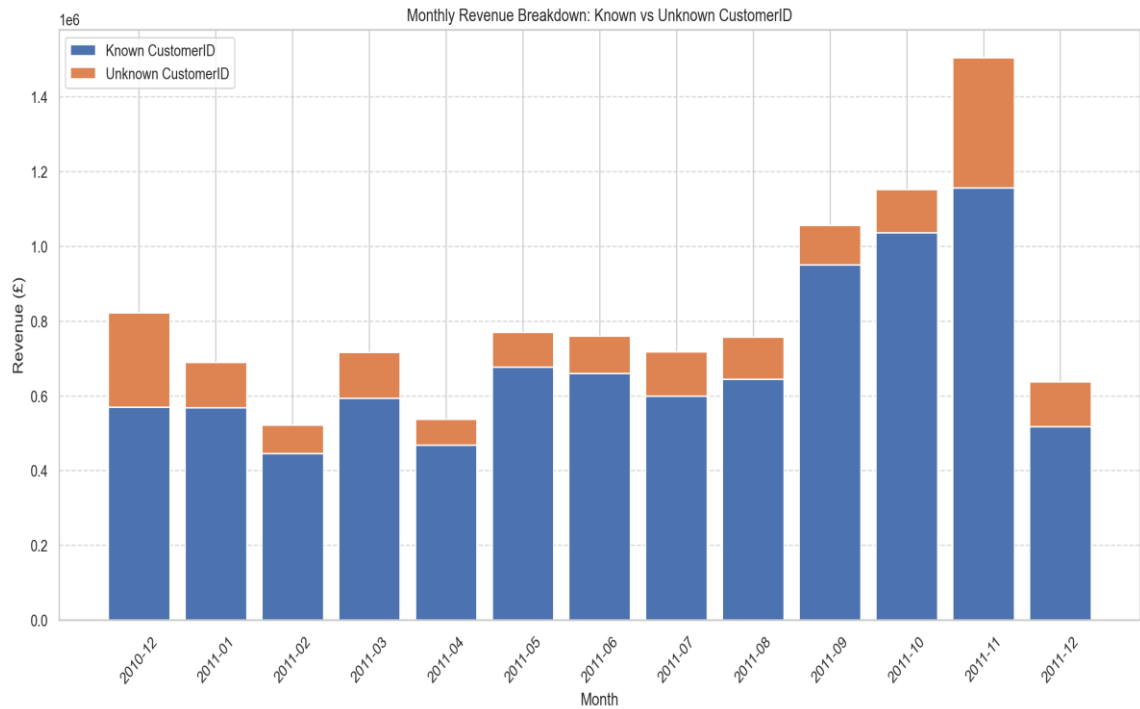
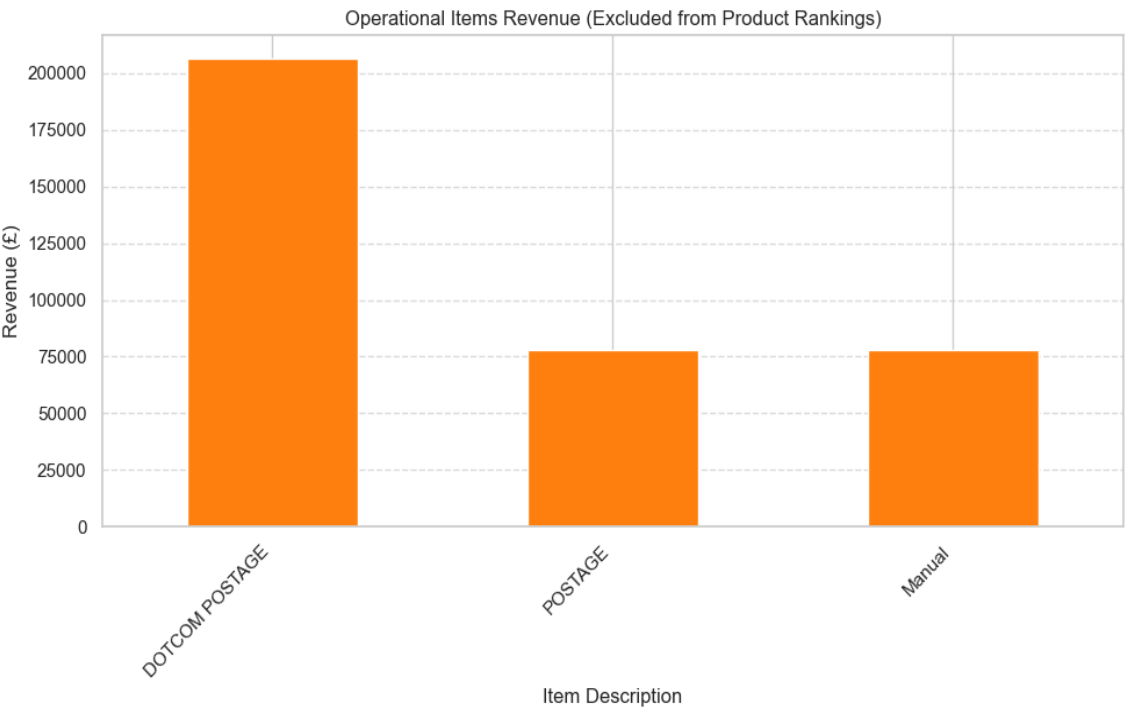
Top product: REGENCY CAKESTAND 3 TIER
£174,156.54 across 1,988 invoices.

Top 3 physical products account for approximately 3.8% of clean physical-product revenue.

Low-revenue tail: candidates for product/catalog review.
The dataset does not contain inventory cost or storage-cost data.

Operational Revenue & Unknown Customer Activity

Separating merchandise performance from non-product revenue and unidentified customer activity



£362K

Operational revenue

16.5%

Revenue linked to Unknown CustomerID

£1.75M

Revenue from unidentified customer transactions

Recommended Action Areas

Actions are framed as testable business responses to observed evidence

1. CUSTOMER RETENTION

Segment the 1,188 Potential Loyalists / At Risk customers for targeted retention tests.

Use the historical £689K revenue baseline as context, not as a guaranteed recovery forecast.

Compare campaign performance against a control group before scaling.

2. SEASONAL PLANNING

Use the November product-mix shift to support seasonal planning.

Monitor RABBIT NIGHT LIGHT and the top seasonal-lift products before the next peak period.

Separate recurring high-volume demand from products whose performance is concentrated in November.

3. CONCENTRATION & PORTFOLIO REVIEW

Monitor UK concentration: 84.6% of revenue and 1,119 Champions.

Evaluate international markets using both revenue and customer counts, especially where sample sizes are small.

Review the low-revenue product tail before making catalog or inventory decisions.

Next Steps for Management

The current analysis identifies where deeper analysis should begin

01 Retention testing

Measure campaign lift among Potential Loyalists / At Risk customers.

02 Customer lifetime value

Estimate the long-term value of Champions and Loyal Customers.

03 Cohort retention

Measure customer retention by acquisition period and identify changing patterns.

04 Market basket analysis

Identify products frequently purchased together for cross-selling analysis.

05 Unknown customer analysis

Investigate the origin and conversion potential of transactions without CustomerID.

Thank You

Questions & Discussion

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UCI Online Retail Dataset — ID 352

*Analysis conducted using Python, Pandas, Matplotlib and Seaborn.
Implementation assistance: Qwen,ai / ChatGPT .*